# Using the theory of subspaces to study matrices.

**Video 12.2.1**

In this final topic we will understand matrices more deeply by studying three important subspaces we can associate to a matrix:

1. The *row space* of a matrix.
2. The *column space* of a matrix.
3. The *null space* of a matrix.

## Row space of a matrix.

First, in words:

The row space of an $m \times n$ matrix $\boldsymbol{M}$ is the subspace of $\mathbb{R}^n$ consisting of all possible vectors you can make by taking linear combinations of the rows of $\boldsymbol{M}$.

***Definition ('Row space of a matrix').***

Consider an $m \times n$ matrix $\boldsymbol{M}$.

Let the vectors in $\mathbb{R}^n$ corresponding to the rows of $\boldsymbol{M}$ be denoted $\vec{r}_1, \dots, \vec{r}_m \in \mathbb{R}^n$.

The row space of $\boldsymbol{M}$, denoted $R(\boldsymbol{M})$, is defined to be the subspace of $\mathbb{R}^n$ defined by

$$R(\boldsymbol{M}) = span\{\vec{r}_1, \dots, \vec{r}_m\} \subset \mathbb{R}^n.$$

$$m\left\{\overbrace{\begin{bmatrix} \leftarrow \vec{r}_1 \rightarrow \\ \leftarrow \vec{r}_2 \rightarrow \\ \vdots \\ \leftarrow \vec{r}_m \rightarrow \end{bmatrix}}^{n}\right.$$

Exercise:

List at least five different vectors in the row space of the following matrix:

$$\begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 1 & 1 \\ -2 & -3 & -4 & -5 \end{bmatrix}.$$

This is a 3x4 matrix.

The three rows are $(1,0,-2,3)$, $(0,1,1,1)$, $(-2,-3,-4,-5) \in \mathbb{R}^4$

$R(M) \subset \mathbb{R}^4$

$1(1,0,-2,3) - 1(0,1,1,1)$

$10(1,0,-2,3) + 100(0,1,1,1) + 1000(-2,-3,-4,-5)$ $\Rightarrow$ elements in the row space

$(0,0,0,0)$

$(1,0,-2,3)$

* the original vectors themselves are elements in the row space

## *Finding a basis for a row space.*

**Video 12.2.2**

- One of the things that will be important in this theory will be the *dimension of a row space*.
- So we need a way to calculate it.
- To calculate dimension we know we have to first find a basis, and then the dimension is just the number of vectors in the basis.
- So the very first thing we need to learn is how to find a basis for a row space.
- We have already met some algorithms for finding bases of subspaces and we could totally just use them here. But there are also some theory that is special to this case that we will learn about.

The algorithm is extremely simple:

<u>The Algorithm to find a basis for the row space of matrix $\boldsymbol{M}$:</u>

To find a basis you

1. Do Gaussian elimination on $\boldsymbol{M}$ to replace $\boldsymbol{M}$ with a row equivalent matrix $\boldsymbol{M}'$ in row echelon form.
2. The non-zero rows of $\boldsymbol{M}'$ form a basis for the row space of $\boldsymbol{M}$.

Method 2 for Example below

$R(M) = \text{span}\{(1,0,1),(2,4,6),(1,4,5),(-4,-8,-12)\}$

$$\begin{bmatrix} 1 & 2 & 1 & -4 \\ 0 & 4 & 4 & -8 \\ 1 & 6 & 5 & -12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & -4 \\ 0 & 4 & 4 & -8 \\ 0 & 4 & 4 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & -4 \\ 0 & 4 & 4 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

contains leading entries

$\therefore$ Basis is $(1,0,1), (2,4,6)$

→ different ans, but must have same number of vectors, which is 2 in this case.

☆ $\dim(R(M)) = 2$

<u>Example:</u>

What is a basis for the row space of the matrix $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 4 & 6 \\ 1 & 4 & 5 \\ -4 & -8 & -12 \end{bmatrix}$?

<u>Solution:</u>

First do Gaussian elimination to put in row echelon form:

don't need to do Gauss Jordan

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 4 & 6 \\ 1 & 4 & 5 \\ -4 & -8 & -12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 4 & 4 \\ 0 & 4 & 4 \\ 0 & -8 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

2 non-zero rows

The answer is the list of non-zero rows of this matrix.

Summary:

the row space of $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 4 & 6 \\ 1 & 4 & 5 \\ -4 & -8 & -12 \end{bmatrix}$ has a basis: $(1,0,1), (0,4,4)$.

**Video 12.2.3**

This algorithm to find a basis for the row space of a matrix is extremely simple but why does it work?

It works because of two relatively simple facts we will think about now:

*Fact #1:*
The result of an elementary row operation has the same row space as the original matrix. → elementary row operation has no effect on row space

*Fact #2:*
If a matrix is in row echelon form then the list of all non-zero rows is already a basis for the row space.

If these two facts are true then it is clear why the algorithm works:

We start with a matrix $\boldsymbol{M}$. Then we do Gaussian elimination to turn it into a row equivalent matrix in row echelon form $\boldsymbol{M}'$ :

*A sequence of elementary row operations given by Gaussian elimination.*

*The given matrix.*

$$\boldsymbol{M} \to \underbrace{\boldsymbol{M}_1 \to \cdots \to \boldsymbol{M}_k}_{} \to \boldsymbol{M}'$$

→ all of them have the same row space.

*The final matrix in row echelon form.*

Then:

$$Basis\ of\ R(\boldsymbol{M}) = Basis\ of\ R(\boldsymbol{M}') = (Nonzero\ rows\ of\ \mathbf{M}').$$

*Fact #1.* *Fact #2.*

Discussion of Fact #1:

The following is also an important reason that row spaces are interesting and useful in the theory of matrices:

***Theorem ('Row equivalent matrices have the same row space').***

Consider two $m \times n$ matrices $\boldsymbol{M}$ and $\boldsymbol{M}'$.

Assume that these two matrices are row equivalent. (Recall that this means we can get from $\boldsymbol{M}$ to $\boldsymbol{M}'$ via a sequence of elementary row operations.)

Then:
the two matrices have the same row space:

$$R(\boldsymbol{M}) = R(\boldsymbol{M}') \subset \mathbb{R}^n.$$

Illustration:

Take the matrix from earlier:

$$\begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 1 & 1 \\ -2 & -3 & -4 & -5 \end{bmatrix}.$$

If we perform Gaussian elimination then we get:

$$\begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 1 & 1 \\ -2 & -3 & -4 & -5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & -3 & -8 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -5 & 4 \end{bmatrix}.$$

These matrices are all row equivalent.
So they have *exactly the same row space.*

*Proof:*

First note that it will be enough to check that row space is unchanged by the three elementary row operations.

(Why? Because if two matrices are row equivalent, then they are related by a sequence of elementary row operations:

$$\boldsymbol{M} \to \boldsymbol{M}_1 \to \cdots \to \boldsymbol{M}_k \to \boldsymbol{M}'.$$

If we know that row space doesn't change for every step in this sequence, then obviously $R(\boldsymbol{M}) = R(\boldsymbol{M}')$.)

So we have three statements to explain:

1. If $\boldsymbol{M}'$ is obtained from $\boldsymbol{M}$ by an elementary row operation of the form $R_i \to R_i + cR_j$ then $R(\boldsymbol{M}) = R(\boldsymbol{M}')$.
2. If $\boldsymbol{M}'$ is obtained from $\boldsymbol{M}$ by an elementary row operation of the form $R_i \leftrightarrow R_j$ then $R(\boldsymbol{M}) = R(\boldsymbol{M}')$.
3. If $\boldsymbol{M}'$ is obtained from $\boldsymbol{M}$ by an elementary row operation of the form $R_i \to cR_i$ then $R(\boldsymbol{M}) = R(\boldsymbol{M}')$.

*Proof (continued):*

In each of the three cases we are going to have to prove that two row spaces are equal: $R(\boldsymbol{M}) = R(\boldsymbol{M}')$.

*How can we do that?*

We can do it in two steps. Let

- the rows of $\boldsymbol{M}$ be denoted $\boldsymbol{R}_1, \dots, \boldsymbol{R}_m \subset \mathbb{R}^n$.
- the rows of $\boldsymbol{M}'$ be denoted $\boldsymbol{R'}_1, \dots, \boldsymbol{R'}_m \subset \mathbb{R}^n$.

*Step 1.* If we can show that each of the vectors $\boldsymbol{R}_1, \dots, \boldsymbol{R}_m$ is a linear combination of the vectors $\boldsymbol{R'}_1, \dots, \boldsymbol{R'}_m$ then

$$R(\boldsymbol{M}) = span\{\boldsymbol{R}_1, \dots, \boldsymbol{R}_m\} \subset span\{\boldsymbol{R'}_1, \dots, \boldsymbol{R'}_m\} = R(\boldsymbol{M}').$$

*Step 2.* The same thing in the other direction. If we can show that each of the vectors $\boldsymbol{R'}_1, \dots, \boldsymbol{R'}_m$ is a linear combination of the vectors $\boldsymbol{R}_1, \dots, \boldsymbol{R}_m$ then:

$$R(\boldsymbol{M}') = span\{\boldsymbol{R'}_1, \dots, \boldsymbol{R'}_m\} \subset span\{\boldsymbol{R}_1, \dots, \boldsymbol{R}_m\} = R(\boldsymbol{M}).$$

Then because $R(\boldsymbol{M}) \subset R(\boldsymbol{M}')$ and $R(\boldsymbol{M}') \subset R(\boldsymbol{M})$ we will know that $R(\boldsymbol{M}) = R(\boldsymbol{M}')$.

to prove equality of sets

*Proof (continued):*

*Case 1:* Assume $\boldsymbol{M'}$ is obtained from $\boldsymbol{M}$ by the $R_i \to R_i + cR_j$.

The rows of $\boldsymbol{M}$ are:

$$\boldsymbol{R}_1, \dots, \boldsymbol{R}_i, \dots, \boldsymbol{R}_j, \dots, \boldsymbol{R}_m$$

and the rows of $\boldsymbol{M'}$ are:

$$\boldsymbol{R'}_1 = \boldsymbol{R}_1, \dots, \boldsymbol{R}'_i = \boldsymbol{R}_i + c\boldsymbol{R}_j, \dots, \boldsymbol{R}'_j = \boldsymbol{R}_j, \dots \boldsymbol{R'}_m = \boldsymbol{R}_m.$$

First note that one direction is easy: we have already just shown how to write the rows of $\boldsymbol{M'}$ as a linear combination of the rows of $\boldsymbol{M}$.

We also need to show the other direction: to write the rows of $\boldsymbol{M}$ as a linear combination of the rows of $\boldsymbol{M'}$. We can see how to do this with a quick calculation and get:

$$\boldsymbol{R}_1 = \boldsymbol{R}'_1, \dots, \boldsymbol{R}_i = \boldsymbol{R}'_i - c\boldsymbol{R}'_j, \dots, \boldsymbol{R}_j = \boldsymbol{R}'_j, \dots, \boldsymbol{R}_m = \boldsymbol{R}'_m.$$

*Proof (continued):*

Here is an illustration of this case 1:

$$\begin{aligned} R'_1 &= R_1 \\ &\vdots \\ R'_i &= R_i + cR_j \\ &\vdots \\ R'_j &= R_j \\ R'_m &= R_m \end{aligned}$$

$$\begin{bmatrix} \leftarrow & \boldsymbol{R}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_i & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_j & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_m & \rightarrow \end{bmatrix} \quad \begin{matrix} \Longrightarrow \\ \Longleftarrow \end{matrix} \quad \begin{bmatrix} \leftarrow & \boldsymbol{R'}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_i & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_j & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_m & \rightarrow \end{bmatrix}$$

$$\begin{aligned} R_1 &= R'_1 \\ &\vdots \\ R_i &= R'_i - cR'_j \\ &\vdots \\ R_j &= R'_j \\ R_m &= R'_m \end{aligned}$$

must show every vector on list to left is a linear combination of the vectors on the right

$$\vec{R}_i = 1(\underbrace{\vec{R}_i + c\vec{R}_j}_{R_i'}) + (-c)\underbrace{\vec{R}_j}_{R_j'}$$

## *Proof (continued):*

For the other cases we'll just give a quick illustration.
Here is case 2:

$$\begin{bmatrix} \leftarrow & \boldsymbol{R}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_i & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_j & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_m & \rightarrow \end{bmatrix} \underset{\begin{array}{rcl} R_1 & = & R'_1 \\ & \vdots & \\ R_i & = & R'_j \\ & \vdots & \\ R_j & = & R'_i \\ R_m & = & R'_m \end{array}}{\overset{\begin{array}{rcl} R'_1 & = & R_1 \\ & \vdots & \\ R'_i & = & R_j \\ & \vdots & \\ R_j' & = & R_i \\ R_m' & = & R_m \end{array}}{\rightleftarrows}} \begin{bmatrix} \leftarrow & \boldsymbol{R'}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_i & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_j & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_m & \rightarrow \end{bmatrix}$$

## *Proof (continued):*

And here is case 3:

$$\begin{bmatrix} \leftarrow & \boldsymbol{R}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_i & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R}_m & \rightarrow \end{bmatrix} \underset{\begin{array}{rcl} R_1 & = & R'_1 \\ & \vdots & \\ R_i & = & \frac{1}{c}R'_i \\ & \vdots & \\ R_m & = & R'_m \end{array}}{\overset{\begin{array}{rcl} R'_1 & = & R_1 \\ & \vdots & \\ R'_i & = & cR_i \\ & \vdots & \\ R_m' & = & R_m \end{array}}{\rightleftarrows}} \begin{bmatrix} \leftarrow & \boldsymbol{R'}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_i & \rightarrow \\ & \vdots & \\ \leftarrow & \boldsymbol{R'}_m & \rightarrow \end{bmatrix}$$

This establishes Fact #1: that row equivalent matrices have the same row spaces.

$\square$

Discussion of Fact #2:

Video 12.2.4

***Fact.***

Consider an $m \times n$ matrix $\boldsymbol{M}$.

Assume $M$ is in row echelon form.

Then:

The list of non-zero rows of $\boldsymbol{M}$ immediately form a basis for the row space of $\boldsymbol{M}$.

Discussion of Fact #2:

We'll understand this fact here by working through a particular example. A careful proof will be left as an exercise for students.

Example:

Consider the following matrix and assume all the variables are non-zero. Observe this matrix is in row echelon form:

$$\boldsymbol{M} = \begin{bmatrix} a & b & c & d & e \\ 0 & 0 & f & g & h \\ 0 & 0 & 0 & i & j \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The algorithm says that a basis for the row space is:

$$(a, b, c, d, e), (0,0, f, g, h), (0,0,0, i, j).$$

Let's understand why this is true directly.

Example (continued):

The definition of row space directly says that the row space is the subspace spanned by the rows:

$R(M) = span\{(a, b, c, d, e), (0,0, f, g, h), (0,0,0, i, j), (0,0,0,0,0)\}$.

We just need the vectors from the non-zero rows. (Because the zero vector here is redundant: it is immediately a linearly combination of the other rows.) So:

$$R(M) = span\{(a, b, c, d, e), (0,0, f, g, h), (0,0,0, i, j)\ \}.$$

To see this list is a basis as always there are two properties to check:

1. The vectors span the subspace. (This is obvious.)
2. The list of vectors is linearly independent. (This we will check.)

Example (continued):

We are currently checking whether the list

$$(a, b, c, d, e), (0,0, f, h, j), (0,0,0, i, j)$$

is linearly independent.

To do that we need to show that the only solution to the following equation $(\star)$

$$c_1(a, b, c, d, e) + c_2(0,0, f, g, h) + c_3(0,0,0, i, j) = (0,0,0,0,0)$$

is the trivial solution $c_1 = c_2 = c_3 = 0$.

There are five equations: one for each entry in this vector. You can check that the augmented matrix corresponding to this system is just the transpose of the original matrix:

$$\left[\begin{array}{ccc|c} a & 0 & 0 & 0 \\ b & 0 & 0 & 0 \\ c & f & 0 & 0 \\ d & g & i & 0 \\ e & h & j & 0 \end{array}\right].$$

<u>Example (continued):</u>

The augmented matrix we are solving is shown below. Recall the columns correspond to the variables $c_1, c_2, c_3$. We can solve this directly, from top to bottom.

*This equation gives:* $c_1 = 0$.

*Using* $c_1 = 0$ *this equation gives:* $c_2 = 0$.

*Using* $c_1 = 0$ *and* $c_2 = 0$ *this equation gives:* $c_3 = 0$.

$$\begin{array}{c} \begin{array}{cccc} c_1 & c_2 & c_3 & \phantom{0} \end{array} \\ \left[\begin{array}{ccc|c} a & 0 & 0 & 0 \\ b & 0 & 0 & 0 \\ c & f & 0 & 0 \\ d & g & i & 0 \\ e & h & j & 0 \end{array}\right] \end{array}.$$

So the only solution to $(\star)$ is $c_1 = c_2 = c_3 = 0$.
So the three vectors are linearly independent.
So this list of three vectors is indeed a basis for the row space of $\boldsymbol{M}$.

$\square$

How could we give a rigorous proof of this that covers all examples?

<u>Exercise</u>

Consider a list of non-zero vectors $\vec{v}_1, \dots, \vec{v}_k \in \mathbb{R}^m$.

Let $n_i$ denote the position of the first non-zero entry of $\boldsymbol{v}_i$.

Assume that if $j > i$ then $n_j > n_i$.

By doing induction on $k$ prove that the list $\vec{v}_1, \dots, \vec{v}_k$ is linearly independent.

(This exercise is in the "Challenging Problems" section of the final tutorial list.)

Video 12.2.5

# Column space of a matrix.

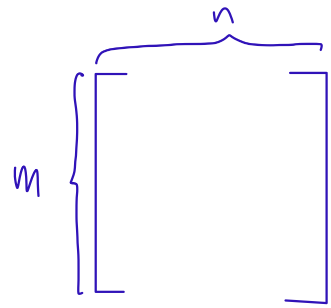
***Definition ('Column space of a matrix').***

Consider an $m \times n$ matrix $\boldsymbol{M}$.

Let the vectors in $\mathbb{R}^m$ corresponding to the columns of $\boldsymbol{M}$ be denoted $\vec{c}_1, \dots, \vec{c}_n \in \mathbb{R}^m$.

The column space of $\boldsymbol{M}$, denoted $C(\boldsymbol{M})$, is defined to be the subspace of $\mathbb{R}^m$ defined by

$$C(\boldsymbol{M}) = span\{\vec{c}_1, \dots, \vec{c}_n\} \subset \mathbb{R}^m.$$




column space is the subspace you got by taking all of the possible combinations of the columns.

column space is a subspace of $\mathbb{R}^m$

Exercise:

List at least five different vectors in the column space of the following matrix:

$$\begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 1 & 1 \\ -2 & -3 & -4 & -5 \end{bmatrix}.$$

$(1,0,-2)$, $10(1,0,-2) - \frac{1}{2}(3,1,5)$, $(1,0,-2) + 10(0,1,3) + 100(-2,1,-4) + 1000(3,1,5)$,

$(0,0,0)$

## *Finding a basis for the column space of a matrix.*

This algorithm is again quite simple. It looks like the algorithm to find a basis for the row space but note that there are some key differences. First we'll meet the algorithm and then we'll think about why it works.

<u>The Algorithm to find a basis for the column space of a matrix $\boldsymbol{M}$:</u>

To find a basis you

1. Do Gaussian elimination on $\boldsymbol{M}$ to replace $\boldsymbol{M}$ with a row equivalent matrix $\boldsymbol{M'}$ in row echelon form.
2. A basis is formed by taking the columns of the original matrix $\boldsymbol{M}$ which have the same position as columns of the row echelon matrix $\boldsymbol{M'}$ which contain leading entries.

<u>Example:</u>

What is a basis for the column space of the following matrix:

$$\boldsymbol{M} = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & -2 & 0 & -1 \end{bmatrix} ?$$

<u>Solution:</u>

The first step in the algorithm is to apply Gaussian elimination to replace $\boldsymbol{M}$ with a row echelon form matrix $\boldsymbol{M'}$ :

$$\boldsymbol{M} \to \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 3/2 & -3 & 3/2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -3/2 & 0 & -3/2 \end{bmatrix} \to \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 3/2 & -3 & 3/2 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \boldsymbol{M'}.$$

Solution (continued):

The second step is to take the columns of the original matrix $\boldsymbol{M}$ which correspond to the columns of $\boldsymbol{M}'$ containing leading entries.

*… so our answer is the corresponding columns of the original matrix.*

*The input matrix $\boldsymbol{M}$.*

$$\begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & -2 & 0 & -1 \end{bmatrix}$$

$$\uparrow \qquad\qquad \uparrow \quad \uparrow$$

*The matrix after Gaussian elimination $\boldsymbol{M}'$.*

$$\begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 3/2 & -3 & 3/2 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

*These are the columns containing leading entries…*

Solution (conclusion):

The column space $C(\boldsymbol{M}) \subset \mathbb{R}^4$ of the matrix

$$\boldsymbol{M} = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & -2 & 0 & -1 \end{bmatrix}$$

has a basis consisting of its first, third, and fourth columns:

$$(2,-1,0,1), (-1,2,1,-2), (0,-3,1,0).$$

Find a basis for the column space of $A = \begin{bmatrix} 1 & -1 & 3 & -1 & 2 \\ 2 & 2 & 1 & -1 & 1 \\ 1 & 0 & 1 & -15 & -1 \\ 1 & 0 & 2 & 4 & 2 \end{bmatrix}$

$$\rightarrow \begin{bmatrix} 1 & -1 & 3 & -1 & 2 \\ 0 & 4 & -5 & 1 & -3 \\ 0 & 1 & -2 & -14 & -3 \\ 0 & 1 & -1 & 5 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 3 & -1 & 2 \\ 0 & 1 & -1 & 5 & 0 \\ 0 & 4 & -5 & 1 & -3 \\ 0 & 1 & -2 & -14 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 3 & -1 & 2 \\ 0 & 1 & -1 & 5 & 0 \\ 0 & 0 & -1 & -19 & -3 \\ 0 & 0 & -1 & -19 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 3 & -1 & 2 \\ 0 & 1 & -1 & 5 & 0 \\ 0 & 0 & -1 & -19 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\square$

* columns containing leading entries will always be the same

Columns 1, 2, and 3 will form the column space

Explanation: So why does this work???

**Video 12.2.7**

Consider a general $m \times n$ matrix organized into columns:

$$\boldsymbol{M} = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{c}_1 & \vec{c}_2 & \cdots & \vec{c}_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}.$$

The definition of column space says immediately that the column space is the span of the columns:

$$C(\boldsymbol{M}) = span\{\vec{c}_1, \dots, \vec{c}_n\} \subset \mathbb{R}^m.$$

To reduce this to a basis: we have to find vectors $\vec{c}_i$ on this list which are linear combinations of the other vectors and then we remove them until we can't find such vectors anymore.

To understand which vectors on this list are linear combinations of the other vectors we have to understand the solutions to the linear dependence equation:

$$a_1\vec{c}_1 + \cdots + a_n\vec{c}_n = (0, \dots, 0) \quad (\star).$$

Exercise

Understand that the augmented matrix of this system of equations

$$a_1\vec{c}_1 + \cdots + a_n\vec{c}_n = (0, \dots, 0) \quad (\star)$$

is the matrix we started with $\boldsymbol{M} = \left[\begin{array}{cccc|c} \uparrow & \uparrow & & \uparrow & 0 \\ \vec{c}_1 & \vec{c}_2 & \cdots & \vec{c}_n & 0 \\ \downarrow & \downarrow & & \downarrow & 0 \end{array}\right].$

augmented matrix → have column of zeros

So: to solve the linear independence equation

$$a_1\vec{c}_1 + \cdots + a_n\vec{c}_n = (0, \dots, 0) \quad (\star)$$

we start by applying Gaussian elimination to $\boldsymbol{M}$:

$$\boldsymbol{M} \rightsquigarrow \boldsymbol{M}'$$

and then interpret the resulting matrix $\boldsymbol{M}'$ as normal.

Explanation (continued).

As usual, after Gaussian elimination the variables split into two types:

- Variables for columns with leading entries $a_{i_1}, \dots, a_{i_k}$.
- Variables for columns without leading entries $a_{j_1}, \dots, a_{j_l}$.

And the general solution will then be of the form:

$$\left\{\begin{array}{l} a_{j_1} = r_1 \\ \quad\vdots \\ a_{j_l} = r_l \\ a_{i_1} = f_1(r_1, \dots, r_l) \\ \quad\vdots \\ a_{i_k} = f_k(r_1, \dots, r_l) \end{array}\right.$$

*As usual, variables from columns of $\boldsymbol{M}'$ without leading entries are set to free parameters…*

*… while variables from columns of $\boldsymbol{M}'$ with leading entries become linear functions of those new free parameters.*

(To be precise, here "linear function" means $f_i(r_1, \dots, r_l) = s_{i1}r_1 + \dots + s_{il}r_l$ for some constants $s_{ij}$. Compare the lecture on determinants for this language.)

Example

$$M = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & -2 & 0 & -1 \end{bmatrix}$$

$$a_1\begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + a_2\begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + a_3\begin{bmatrix} -1 \\ 2 \\ 1 \\ -2 \end{bmatrix} + a_4\begin{bmatrix} 0 \\ -3 \\ 1 \\ 0 \end{bmatrix} + a_5\begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccccc|c} 2 & 2 & 1 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccccc|c} -1 & -1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 3 & -6 & 3 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \end{array}\right]$$

$$\to \left[\begin{array}{ccccc|c} -1 & -1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$a_2 = r$  $a_5 = s$

$a_4 = 0$

$a_3 = -s$

$a_1 = -r + 2(-s) - 3 \cdot 0 + s = -r - s$

basis

$$(-r-s)\begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + r\begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + (-s)\begin{bmatrix} -1 \\ 2 \\ 1 \\ -2 \end{bmatrix} + (0)\begin{bmatrix} 0 \\ -3 \\ 1 \\ 0 \end{bmatrix} + s\begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

columns in REF without leading entries (became free parameters)

set $r=1, s=0$,

$$1\begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} = 1\begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

set $r=0, s=1$

$$1\begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 1 \\ -2 \end{bmatrix}$$

$\therefore$ we eliminate these 2 to form the basis

Explanation (continued).

Now we just have to interpret this general solution.
(Keep in mind that this is a parameterization for the set of all possible solutions of the linear independence equation $(\star)$.)

It is easy but a bit technical to deduce from this general solution that…

$$\left\{\begin{array}{l} a_{i_1} = f_1(r_1, \dots, r_l) \\ \quad\vdots \\ a_{i_k} = f_k(r_1, \dots, r_l) \\ a_{j_1} = r_1 \\ \quad\vdots \\ a_{j_l} = r_l \end{array}\right.$$

*…the columns of the original $\boldsymbol{M}$ $\vec{c}_{i_1}, \dots, \vec{c}_{i_k}$ corresponding to these columns of $\boldsymbol{M}'$ containing leading entries are linearly independent…*

*… while the other columns of the original $\boldsymbol{M}$ $\vec{c}_{j_1}, \dots, \vec{c}_{j_k}$ corresponding to these columns of $\boldsymbol{M}'$ without leading entries are linear combinations of the first set of columns $\vec{c}_{i_1}, \dots, \vec{c}_{i_k}$.*

Exercise:
Finish the explanation by checking the statements in these two boxes.

## *The rank of a matrix.*

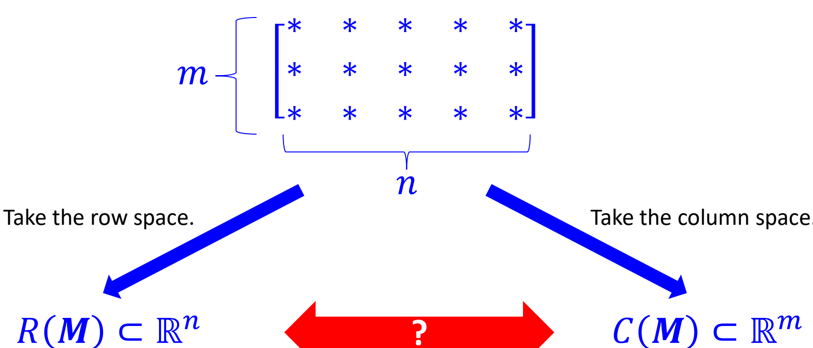
**Video 12.2.8**

Next we are going to introduce the rank of a matrix $\boldsymbol{M}$.
This is an important non-negative integer associated to any matrix.

$$\operatorname{rank}(M) \in \mathbb{Z}_{\geq 0}.$$

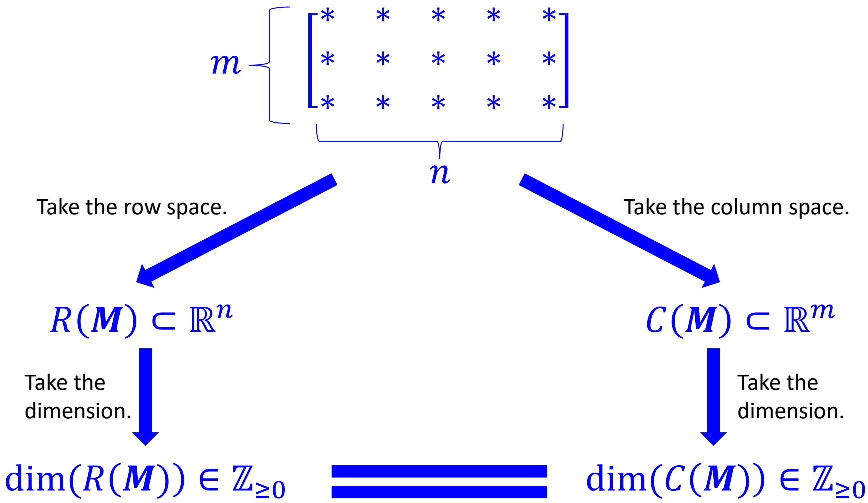
Here is the quick story of *rank*.

Consider an $m \times n$ matrix $M$. We have just met two subspaces we can associate with $\boldsymbol{M}$. Note that they live in different Euclidean spaces.




Is there any relationship between these two subspaces???

It would be surprising if there was…
after all $R(\boldsymbol{M})$ lives in $\mathbb{R}^n$ while $C(\boldsymbol{M})$ lives in the possibly different $\mathbb{R}^m$…




It turns out these two dimensions are **equal.**
We will prove this surprising fact shortly.
This special number will be called the **rank** of $\boldsymbol{M}$.

***<u>Theorem and definition ('Rank of a matrix').</u>***

Consider an $m \times n$ matrix $\boldsymbol{M}$.

Then:

$$\dim(R(\boldsymbol{M})) = \dim(C(\boldsymbol{M})) .$$

This number is called the **rank** of $\boldsymbol{M}$.

We have to explain why this is true. (i.e. prove it.)

The explanation is quite simple actually once you have understood from earlier how to find bases for the row space and column space.

Example:

Consider the following matrix $\boldsymbol{M} = \begin{bmatrix} 1 & -4 & 2 & 6 & 0 \\ 3 & 2 & 0 & -1 & 2 \\ -1 & -10 & 4 & 13 & -2 \end{bmatrix}$.

Calculate:

1. A basis for the row space.
2. A basis for the column space.
3. Check that indeed the dimension of the row space of $\boldsymbol{M}$ equals the dimension of the column space of $\boldsymbol{M}$ like we are claiming.

Solution:

Both the algorithms we need start by putting the matrix in row echelon form via Gaussian elimination:

$$\boldsymbol{M} \to \begin{bmatrix} 1 & -4 & 2 & 6 & 0 \\ 0 & 14 & -6 & -19 & 2 \\ 0 & -14 & 6 & 19 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & -4 & 2 & 6 & 0 \\ 0 & 14 & -6 & -19 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \boldsymbol{M}'.$$

Solution:

We just did Gaussian elimination to put $M$ in row echelon form:

$$\boldsymbol{M} = \begin{bmatrix} 1 & -4 & 2 & 6 & 0 \\ 3 & 2 & 0 & -1 & 2 \\ -1 & -10 & 4 & 13 & -2 \end{bmatrix}$$

Gaussian elimination.

$$\begin{bmatrix} 1 & -4 & 2 & 6 & 0 \\ 0 & 14 & -6 & -19 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Now we can just read off the answers from this row echelon form:

1. A basis for the row space is formed by the non-zero rows of $\boldsymbol{M}'$ marked by the **red** arrows. The basis is: $(1, -4,2,6,0), (0,14, -6, -19,2)$.

2. A basis for the column space is formed by the columns of $\boldsymbol{M}$ corresponding to the columns of $\boldsymbol{M}'$ containing leading entries. See the **green** arrows. The basis is: $(1,3, -1), (-4,2, -10)$.

3. As predicted these two bases have the same number of vectors. The number of **red** arrows always equals the number of **green** arrows. $\square$

Now let's write down a proof that: $\dim(R(\boldsymbol{M})) = \dim(C(\boldsymbol{M}))$

Let $\boldsymbol{M}'$ denote a matrix you get by doing Gaussian elimination on $\boldsymbol{M}$.

$$\boldsymbol{M} \xrightarrow{\text{Gaussian elimination.}} \boldsymbol{M}'$$

A row equivalent matrix in row echelon form.

$\dim(R(\boldsymbol{M}))$

$= (\#\text{ of vectors in a basis for } R(M))$

$= (\#\text{ of non-zero rows of } M')$ Why? → because of the algorithm

every non-zero row has a leading entry

$= (\#\text{ of leading entries in } M')$

$= (\#\text{ of columns of } M' \text{ containing leading entries})$

$= (\#\text{ of vectors in a basis for } C(M))$ Why? → because of the algorithm

$= \dim(C(\boldsymbol{M}))$

Because the non-zero rows of the matrix $\boldsymbol{M}'$ you get after Gaussian elimination form a basis for $R(\boldsymbol{M})$.

$\dim(R(\boldsymbol{M}))$

$= (\#\text{ of vectors in a basis for } R(M))$

$= (\#\text{ of non-zero rows of } M')$

$= (\#\text{ of leading entries in } M')$

$= (\#\text{ of columns of } M' \text{ containing leading entries})$

$= (\#\text{ of vectors in a basis for } C(M))$

$= \dim(C(\boldsymbol{M}))$

Because the columns of $\boldsymbol{M}'$ containing leading entries tell you which columns of $\boldsymbol{M}$ to take to form a basis for $C(\boldsymbol{M})$.

$\square$

## *The Nullspace of a matrix*

Video 12.2.9

This is the third of the three important subspaces attached to a matrix.

Definition: Consider an $m \times n$ matrix $\boldsymbol{M} = [m_{ij}]$.

$$\begin{bmatrix} m_{11} & \cdots & \cdots & \cdots & m_{1n} \\ \vdots & & & & \vdots \\ m_{m1} & & & & m_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ \vdots \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

The null space of $\boldsymbol{M}$, denoted $N(\boldsymbol{M})$, is defined to be the subspace of $\mathbb{R}^n$ consisting of $n$-tuples $(v_1, \dots, v_n)$ which solve this equation.

null space is the set of solutions to a homogeneous system

In words:

The null space of $\boldsymbol{M}$, denoted $N(\boldsymbol{M})$, is the set of vectors $\vec{v}$ which are sent to the zero vector when you multiply them by $\boldsymbol{M}$. That is: $\boldsymbol{M}\vec{v} = \boldsymbol{0}$.

Here is an important related definition:

***Definition ('Nullity of a matrix').***

Consider a matrix $\boldsymbol{M}$.

The nullity of $\boldsymbol{M}$, denoted $\text{nullity}(\boldsymbol{M})$, is defined to be the dimension of the null space of $M$:

$$\text{nullity}(\boldsymbol{M}) = \dim(N(\boldsymbol{M})).$$

## *Finding a basis for the null space.*

Actually we know how to do this already, but let's think it through again.

We have a matrix $\boldsymbol{M} = [m_{ij}]$ and we are trying to solve the following matrix equation for the variables $v_1, \dots, v_n$ :

$$\begin{bmatrix} m_{11} & \cdots & \cdots & \cdots & m_{1n} \\ \vdots & & & & \vdots \\ m_{m1} & & & & m_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ \vdots \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Each side of this equation is an $n \times 1$ matrix.
Equating the two sides of this equation entry-by-entry turns this single matrix equation into a system of $m$ linear equations in the $n$ variables $v_1, \dots, v_n$.

The augmented matrix of this linear system is precisely: $\left[ \boldsymbol{M} \,\middle|\, \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} \right]$.

We can proceed to solve this set of solutions using the techniques we have learned for solving systems of equations.

<u>Example:</u>

Find a basis for the null space and calculate the nullity of the matrix

$$\boldsymbol{M} = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & -2 & 0 & -1 \end{bmatrix}.$$

<u>Solution:</u>

To find the solution space we need to solve the following augmented matrix in the usual way:

$$\left[\begin{array}{ccccc|c} 2 & 2 & -1 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \end{array}\right].$$

We can solve this system by doing Gauss-Jordan elimination to put the matrix into reduced row-echelon form. In this case the result is:

$$\left[\begin{array}{ccccc|c} 2 & 2 & -1 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \end{array}\right] \to \cdots \to \underbrace{\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]}_{\text{RREF}}$$

Solution (continued):

The reduced row echelon form we found was

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

As usual: variables associated to columns without leading entries (here variables $v_2$ and $v_5$) become free variables while columns containing leading entries are solved in terms of these free variables.

The general solution is: $$\begin{cases} v_1 &= -r-s \\ v_2 &= r \\ v_3 &= -s \\ v_4 &= 0 \\ v_5 &= s \end{cases}, \ r, s \in \mathbb{R}.$$ → parameterisation for the null space

In set theory notation:

$$\{(-r-s, r, -s, 0, s); r, s \in \mathbb{R}\} \subset \mathbb{R}^5.$$

Solution (continued):

To find a basis for this subset of $\mathbb{R}^5$

$$\{(-r-s, r, -s, 0, s); r, s \in \mathbb{R}\}$$

we separate variables as usual and write:

$$(-r-s, r, -s, 0, s) = r(-1,1,0,0,0) + s(-1,0,-1,0,1).$$

These two vectors

$$(-1,1,0,0,0), (-1,0,-1,0,1)$$

is thus a spanning set for the null space of $\boldsymbol{M}$.

In addition they are obviously linearly independent.
Thus they form a basis for $N(\boldsymbol{M})$:

$$(-1,1,0,0,0), (-1,0,-1,0,1).$$

Finally: the nullity of $\boldsymbol{M}$ is the dimension of the $N(\boldsymbol{M})$. The basis we have found contains two vector so obviously

$$\text{nullity}(\boldsymbol{M}) = 2.$$

□

*Question.*

We are given a matrix $\boldsymbol{M}$.

We perform Gaussian elimination on $\boldsymbol{M}$ and do a sequence of row operations to turn it into a row equivalent matrix in row echelon form $\boldsymbol{M}'$.

$$\boldsymbol{M} \to \cdots \to \boldsymbol{M}'.$$

How can we know the nullity of $\boldsymbol{M}$ just by looking at the matrix $\boldsymbol{M}'$ ?

*Answer.*

The nullity of $\boldsymbol{M}$ is exactly the number of columns of $\boldsymbol{M}'$ without leading entries.

*Why?*

$$\begin{aligned} \text{nullity}(\boldsymbol{M}) &= (\#\text{ of vectors in a basis for } N(\boldsymbol{M})) \\ &= (\#\text{ of free parameters in the general solution of } [\boldsymbol{M}|\boldsymbol{0}]) \\ &= (\#\text{ of columns of } \boldsymbol{M}' \text{without leading entries}) \end{aligned}$$

**Video 12.2.10**

What is the nulity of $\begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 2 & 4 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 1 & 2 & 1 & -1 & 1 \end{bmatrix}$ ?

$\to \begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\therefore$ Nulity $= 2$

## *The Rank-Nullity theorem.*

**Video 12.2.11**

Consider an $m \times n$ matrix $\boldsymbol{M}$.

We now have a collection of non-negative integers naturally associated to this matrix:

$\text{rank}(\boldsymbol{M})$

$\text{nullity}(\boldsymbol{M})$

$n = (\#\text{ of columns of } \boldsymbol{M})$

$m = (\#\text{ of rows of } \boldsymbol{M})$

Let's try and guess if there is some fixed relationship between these numbers …

*Observation:*

These four numbers associated to any matrix $\boldsymbol{M}$:

- $\operatorname{rank}(\boldsymbol{M})$
- $\operatorname{nullity}(\boldsymbol{M})$
- $m = (\text{\# of rows of } \boldsymbol{M})$
- $n = (\text{\# of columns of } \boldsymbol{M})$

can all be read directly from a matrix $\boldsymbol{M}'$ you get by doing Gaussian elimination to $\boldsymbol{M}$:

$$\boldsymbol{M} \to \cdots \to \boldsymbol{M}'.$$

*How?*

- $\operatorname{rank}(\boldsymbol{M}) = \boxed{\quad ? \quad}$ → (# of columns of M' containing leading entries)
- $\operatorname{nullity}(\boldsymbol{M}) = \boxed{\quad ? \quad}$ → (# of columns of M' without leading entries)

*So what is a relationship between these numbers???*

*Let's try adding rank to nullity…*

$$\underbrace{\operatorname{rank}(\boldsymbol{M})}_{\left(\substack{\textit{Number of columns of } \boldsymbol{M}' \\ \textit{containing leading} \\ \textit{entries}}\right)} + \underbrace{\operatorname{nullity}(\boldsymbol{M})}_{\left(\substack{\textit{Number of columns of } \boldsymbol{M}' \\ \textit{without leading} \\ \textit{entries}}\right)}$$

$$= \textit{Number of columns of } \boldsymbol{M}'.$$

$$= \textit{Number of columns of } \boldsymbol{M}.$$

We have just discovered/proved what is called the ***Rank-Nullity Theorem***.

***Theorem ('Rank-Nullity Theorem').***

Consider an $m \times n$ matrix $\boldsymbol{M}$.

Then:

$$\text{rank}(\boldsymbol{M}) + \text{nullity}(\boldsymbol{M}) = n,$$

the number of columns of $\boldsymbol{M}$.

Example

**Video 12.2.12**

Consider the following matrix:

$$\boldsymbol{M} = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 1 & -2 & -1 & -6 \\ 3 & 2 & -1 & -6 \end{bmatrix}.$$

1. Find a basis for the row space of $\boldsymbol{M}$.
2. Find a basis for the column space of $\boldsymbol{M}$.
3. Find a basis for the null space of $\boldsymbol{M}$.
4. Observe that the Rank-Nullity theorem holds for this example.

(Note: this calculation was on the 2018 final exam for MH1200.)

Solution

All four parts of this problem can be written down directly from a reduced row echelon form of $\boldsymbol{M}$. So we'll start by performing Gauss-Jordan elimination on $\boldsymbol{M}$.

$$\boldsymbol{M}=\begin{bmatrix}1&2&0&3\\1&-2&-1&-6\\3&2&-1&-6\end{bmatrix}\to\begin{bmatrix}1&2&0&3\\0&-4&-1&-9\\0&-4&-1&-15\end{bmatrix}\to\begin{bmatrix}1&2&0&3\\0&-4&-1&-9\\0&0&0&-6\end{bmatrix}$$

$$\to\begin{bmatrix}1&2&0&0\\0&-4&-1&0\\0&0&0&1\end{bmatrix}\to\underbrace{\begin{bmatrix}1&0&-1/2&0\\0&1&1/4&0\\0&0&0&1\end{bmatrix}}_{\text{RREF}}=\boldsymbol{M}'$$

Solution

We can now read of the various bases associated to $\boldsymbol{M}=\begin{bmatrix}1&2&0&3\\1&-2&-1&-6\\3&2&-1&-6\end{bmatrix}$ from $\boldsymbol{M}'$ the reduced row echelon form of $\boldsymbol{M}$ :

$$\begin{matrix}\rightarrow\\\rightarrow\\\rightarrow\end{matrix}\begin{bmatrix}1&0&-1/2&0\\0&1&1/4&0\\0&0&0&1\end{bmatrix}$$

*1. Basis for the row space $R(\boldsymbol{M})$:*

The non-zero rows of $\boldsymbol{M}'$ form a basis for the row space of $\boldsymbol{M}$.
Thus a basis for $R(\boldsymbol{M})$ is given by:

$$\left(1,0,-\frac{1}{2},0\right),\left(0,1,\frac{1}{4},0\right),(0,0,0,1).\ \Rightarrow \text{implies rank}(M)=3$$

Solution (continued)

We can now read of the various bases associated to $\boldsymbol{M} = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 1 & -2 & -1 & -6 \\ 3 & 2 & -1 & -6 \end{bmatrix}$

from $\boldsymbol{M}'$ the reduced row echelon form of $\boldsymbol{M}$ :

$$\begin{bmatrix} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 1/4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

*2. Basis for the column space $C(\boldsymbol{M})$:*

The columns of $\boldsymbol{M}'$ containing a leading entry tell you which columns of the original matrix $\boldsymbol{M}$ to take for the basis.

Thus a basis for $C(\boldsymbol{M})$ is given by:

$$(1,1,3), (2,-2,2), (3,-6,-6).$$ $\Rightarrow$ implies $\operatorname{rank}(M) = 3$

Solution (continued)

We can now read of the various bases associated to $\boldsymbol{M} = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 1 & -2 & -1 & -6 \\ 3 & 2 & -1 & -6 \end{bmatrix}$

from $\boldsymbol{M}'$ the reduced row echelon form of $\boldsymbol{M}$ :

$$\begin{bmatrix} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 1/4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

*3. Basis for the null space $N(\boldsymbol{M})$:*

There will be one basis vector in the null space for every free parameter in the general solution of this matrix. Thus there will be one basis vector for every column in $\boldsymbol{M}'$ not containing a leading entry. The general solution is:

$N(M) = \{(\frac{r}{2}, -\frac{r}{4}, r, 0) ; r \in \mathbb{R}\} = \operatorname{span}\{(\frac{1}{2}, -\frac{1}{4}, 1, 0)\}$

$$\begin{cases} v_1 &= r/2 \\ v_2 &= -r/4 \\ v_3 &= r \\ v_4 &= 0 \end{cases}$$

. Thus $N(\boldsymbol{M})$ has a basis with one vector: $\left(\frac{1}{2}, -\frac{1}{4}, 1, 0\right)$.

parameterisation for the null space

<u>Solution (continued)</u>

We can now read of the various bases associated to $\boldsymbol{M} = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 1 & -2 & -1 & -6 \\ 3 & 2 & -1 & -6 \end{bmatrix}$

from $\boldsymbol{M}'$ the reduced row echelon form of $\boldsymbol{M}$ :

$$\begin{bmatrix} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 1/4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

***4. Checking the Rank-Nullity theorem.***

rank(***M***)+nullity(***M***)
=# of **green** arrows (3) + # of **orange** arrows (1)
= # of columns (4).

□